

No association between maternal and child PFAS concentrations and repeated measures of ADHD symptoms at age 2½ and 5 years in children from the Odense Child Cohort

Dalsager, Louise; Jensen, Tina Kold; Nielsen, Flemming; Grandjean, Philippe; Bilenberg, Niels; Andersen, Helle Raun

Published in:
Neurotoxicology and Teratology

DOI:
[10.1016/j.ntt.2021.107031](https://doi.org/10.1016/j.ntt.2021.107031)

Publication date:
2021

Document version:
Accepted manuscript

Document license:
CC BY-NC-ND

Citation for pulished version (APA):

Dalsager, L., Jensen, T. K., Nielsen, F., Grandjean, P., Bilenberg, N., & Andersen, H. R. (2021). No association between maternal and child PFAS concentrations and repeated measures of ADHD symptoms at age 2½ and 5 years in children from the Odense Child Cohort. *Neurotoxicology and Teratology*, 88, Article 107031. <https://doi.org/10.1016/j.ntt.2021.107031>

Go to publication entry in University of Southern Denmark's Research Portal

Terms of use

This work is brought to you by the University of Southern Denmark.
Unless otherwise specified it has been shared according to the terms for self-archiving.
If no other license is stated, these terms apply:

- You may download this work for personal use only.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying this open access version

If you believe that this document breaches copyright please contact us providing details and we will investigate your claim.
Please direct all enquiries to puresupport@bib.sdu.dk

No association between maternal and child PFAS concentrations and repeated measures of ADHD symptoms at age 2½ and 5 years in children from the Odense Child Cohort

Louise Dalsager¹, Tina Kold Jensen^{1,2,3}, Flemming Nielsen¹, Philippe Grandjean^{1,4}, Niels Bilenberg⁵, Helle Raun Andersen¹

¹Department of Clinical Pharmacology, Pharmacy and Environmental Medicine, Institute of Public Health, University of Southern Denmark, Odense, Denmark

²Hans Christian Andersen Children's Hospital, Odense University Hospital, Odense, Denmark

³Odense Patient data Explorative Network (OPEN), Odense, Denmark

⁴Department of Environmental Health, Harvard T.H.Chan School of Public Health, Boston, MA 02115, USA

⁵Research center in Neurodevelopmental Disorders (FOCUS), Department of Child and Adolescent Mental Health Odense, Mental Health Services in the Region of Southern Denmark, University of Southern Denmark, Odense, Denmark

Acknowledgements

The technicians at Hans Christian Andersen's Children's Hospital are acknowledged for their careful examination of the children. The lab technicians at the Department of Clinical Pharmacology, Pharmacy and Environmental Medicine, University of Southern Denmark are acknowledged for their thorough work in analyzing the blood samples. This work was supported by Odense University Hospital, Denmark, the Region of Southern Denmark, The Municipality of Odense, Denmark, The University of Southern Denmark, Odense Patient data Explorative Network (OPEN), Denmark, The National Board of Social Services, Denmark, The Mental Health Research Fund of Southern Denmark, the Danish Council for Independent Research, Medical Sciences (4004-00352B and 8020-00123B_FSS), Novo Nordisk Foundation, Denmark (grant no. NNF19OC0058266 and NNF17OC0029404), HBM4EU, EU Horizon 2020 (grant no. 733032). PG is supported by the National Institute for Environmental Health Sciences (ES027706)

1 Abstract

2

3 Introduction: The potential impact of exposure to perfluoroalkyl substances (PFAS) on childhood
4 Attention-Deficit Hyperactivity-Disorder (ADHD) is unclear and deserves scrutiny. The majority of
5 previously conducted longitudinal studies found no association between maternal serum-PFAS
6 concentrations and ADHD symptoms in the offspring, but some studies observed possible
7 associations with postnatal PFAS exposures, mainly in girls.

8

9 Objective: To investigate the association between maternal and child serum concentrations of five
10 PFAS and symptoms of ADHD at ages 2½ and 5 years.

11

12 Methods: In the Odense Child Cohort (OCC) women were recruited in early pregnancy in 2010-12
13 and their children are being prospectively followed. Mothers donated serum samples in the first
14 trimester and children at age 18 months to be analyzed for perfluorohexane sulfonic acid (PFHxS),
15 perfluorooctanesulfonic acid (PFOS), perfluorooctanoic acid (PFOA), perfluorononanoic acid
16 (PFNA) and perfluorodecanoic acid (PFDA). Parents completed the Child Behavior Check List for
17 ages 1.5-5 years (CBCL/1½-5), including a 6-item ADHD symptom scale at age 2½ years and again
18 at 5 years. Negative binomial and logistic regression models taking account of repeated measures
19 were used to investigate the association between maternal and child serum-PFAS concentrations
20 and the ADHD symptom score. Effect modification by child sex was investigated as well.

21

22 Results: A total of 1,138 mother-child pairs were included. At age 2½ years, 17.4 % of the children
23 had an ADHD scale score ≥ 5 (equivalent to the 90th percentile), whereas the proportion was 15.8 %

24 at age 5. We found no association between either maternal or child PFAS concentrations in serum
25 and symptoms of ADHD at age 2½ or 5 years, and no evidence of effect modification by sex.

26

27 Conclusion: We found no evidence of an association between early-life PFAS exposure and the risk
28 of developing symptoms of ADHD.

29

30

31

32 1. Introduction

33 Perfluoroalkyl substances (PFAS) are highly stable chemicals, and due to their grease- and water
34 repelling properties, they are widely used in industrial production of food wrapping materials,
35 carpets, and textiles [1, 2]. PFAS are found in the vast majority of human blood samples analyzed
36 worldwide with the major sources of exposure being intake of contaminated food and drinking
37 water or the use of PFAS-containing consumer products [3]. Due to previously heavy use and long
38 elimination half-lives [4, 5], the most frequently detected PFAS are still perfluorooctane sulfonic
39 acid (PFOS) and perfluorooctanoic acid (PFOA), although they are now restricted under the
40 Stockholm Convention [2]. In pregnant women, the PFAS are transferred through the placenta to
41 the fetus and later via breastmilk to the infant [6, 7]. The negative impact of PFAS on different
42 aspects of human health and development is evident, and they have been proposed to be
43 developmental neurotoxicants [8].

44 One of the most common neurodevelopmental disorders in childhood is Attention-Deficit
45 Hyperactivity Disorder (ADHD), which is characterized by symptoms of inattention, hyperactivity
46 and impulsivity [9]. The etiology of ADHD is largely unknown, but existing literature suggests
47 involvement of both genetic and environmental factors [9, 10]. Several epidemiological studies
48 have investigated the association between early-life exposure to PFAS and development of ADHD
49 in childhood, including a meta-analysis of nine European studies [11], but the findings have been
50 inconsistent [12]. Among longitudinal studies, the majority found no overall association between
51 PFAS concentration measured in maternal serum or umbilical cord blood and ADHD diagnosis or
52 symptoms [13-19]. However, some studies revealed an association between PFAS exposure and
53 ADHD, mainly in girls [20-25]. Common for all studies were a relatively modest sample size, and
54 most had only a single measure of ADHD symptoms in childhood and also only measurement of
55 PFAS-exposure from one timepoint. Thus, the purpose of the present study was to investigate the

56 association between both maternal and child PFAS concentrations in regard to symptoms of ADHD
57 both at age 2½ and 5 years, and whether the association differed according to child sex.

58

59 2. Methods

60 2.1 Study participants and data collection

61 The study was conducted within the Odense Child Cohort (OCC), which is an ongoing cohort study
62 focusing on the health and development of children living in Odense, Denmark. Details on
63 recruitment, data collection and participants have already been published [26]. A total of 2,874
64 pregnant women were enrolled in the cohort during the recruitment period (2010-2012) (Figure 3).

65 At the time of maternal enrollment in the study (GW 8-16), a blood sample was drawn, and
66 repeatedly during pregnancy and after childbirth, the participants filled in questionnaires on e.g.,
67 health, lifestyle and social factors [27]. At age 18 months, the children were invited to donate a
68 blood sample. Information on maternal age, pre-pregnancy body mass index (BMI), smoking during
69 pregnancy, parity, date of childbirth, prematurity, birth weight and child sex was derived from the
70 hospital birth records. Perceived stress during pregnancy (Cohen's PSS10) was reported in
71 questionnaires, and any parental psychiatric diagnoses prior to birth of the index child were
72 retrieved from the Danish National Patient Registry. Information on maternal educational level
73 obtained from the questionnaires was missing for some women for whom occupation was retrieved
74 from the birth records. By combining information from questionnaires and a sub-project on
75 breastfeeding reported via text messages, the total duration of breastfeeding was obtained.

76 The study was performed in accordance with the second Helsinki Declaration and approved by the
77 regional Ethical Review Committee (Project ID S-20090130). All participants received written and
78 oral information and gave their written consent.

79

80 2.2 PFAS measurements

81 Serum samples obtained at inclusion from 1,699 women and at age 18 months from 941 children
82 were analyzed for concentrations of PFHxS, PFOS, PFOA, PFNA and PFDA using online solid-
83 phase extraction followed by liquid chromatography and triple quadrupole mass spectrometry (LC-
84 MS/ MS) at Environmental Medicine, University of Southern Denmark [28] Of the maternal
85 samples, 199 were analyzed in 2011, further 521 in 2013, and the remaining 979 in 2019. All child
86 samples were analyzed in 2020. The within-batch coefficients of variation (CVs) were < 3% and the
87 between batch CVs for all sets analyzed were < 10.5% for maternal samples and < 7% for child
88 samples. Quality control and quality assessment were based on certified reference material
89 (NIST1958) from the National Institute of Standards and Technology and participating in the
90 external quality control program, German External Quality Assessment Scheme (G-EQUAS) for analysis
91 in biological materials organized by the German Society of Occupational Medicine. For this study,
92 PFOS, PFOA, PFNA and PFDA were measurable in all blood samples, whereas six women had a
93 PFHxS concentration below the limit of quantification (LOQ (0.03 ng/ml)) that was replaced by
94 LOQ/2.

95

96 2.3 Child Behavior Checklist 1.5-5 (CBCL/1½-5) – The ADHD symptom score

97 At a mean age of 27 (± 3) months and again at a mean age of 5.4 (± 0.4) years, families participating
98 in the OCC were invited to complete the CBCL/1½-5. For the first examination, an invitation to fill
99 in the questionnaire online and a password was mailed to the parents of 2,536 children (out of the
100 2,551 in the OCC, as 15 did not wish to be contacted), of which 1,942 (76.6 %) filled in the
101 questionnaire. At 5 years of age, the same procedure was used to contact parents of 2,414 children
102 still participating in the OCC, and of these, 1,206 filled in the questionnaire (50.0%).

103 The CBCL/1½-5 is a well-established and frequently used parent checklist with solid psychometric
104 properties [29-31]. As no simple clinical test can determine psychopathology in children before
105 school age, informant reporting is standard practice in this field [32, 33]. The checklist consists of
106 100 questions regarding behavioral, emotional, and social problems, six of which represent an
107 ADHD problem scale (cannot concentrate, hyperactive, cannot wait, demands met immediately,
108 quickly shifts, gets into everything). All items are rated either not true (0), somewhat/sometimes
109 true (1) or very often/often true (2), giving a maximal total score of 12. As ADHD is recognized as
110 a spectrum disorder and children seldom fulfill complete diagnostic criteria at this early age, the
111 responses reflect diagnostic traits. A score on the CBCL/1½-5 ADHD scale equal to or above 5 is a
112 predictor of a later ADHD diagnosis [34].

113

114 2.4 Statistics

115 The serum-PFAS concentrations as well as the child ADHD symptom scores are reported as
116 medians with 95th percentiles due to a non-normal distribution and presented according to
117 participant characteristics. Characteristics included were maternal pre-pregnancy BMI (<25 kg/m²,
118 25-30 kg/m², >30 kg/m²), maternal educational level (high school only, vocational training or less,
119 high school + 1-4 years, high school + 5 or more years), parity (0/ >0), smoking during pregnancy
120 (yes/no), maternal age at birth (<30, 30-34, >34 years), parental psychiatric diagnosis (yes/no),
121 maternal perceived stress score during pregnancy (0-10, 10-15, 16-30, missing), prematurity
122 (gestational age <37 weeks/≥37 weeks), child sex (boy/girl) and duration of breastfeeding (0-23,
123 24-40, 41-141 weeks, missing).

124 As information on breastfeeding was missing for a considerable proportion of participants (12.5 %),
125 we imputed the missing values using multiple imputation by chained equations with 100
126 imputations based on all variables included in the regression model (child PFAS concentrations,

127 child ADHD scale score, child sex, maternal educational level, parity, maternal age at delivery and
128 parental psychiatric diagnosis). The imputations were performed using the mi impute chained
129 command in Stata version 16.1 (StataCorp, College Station, TX). The distribution of the original
130 and the imputed breastfeeding variables were close to identical (<6 % change), and the imputed
131 variable was used in all analyses. Correlations between log transformed maternal and child PFAS
132 concentrations were investigated using Pearson's correlation coefficient (r_p). To investigate the
133 potential impact of loss-to-follow-up, we compared characteristics of those who did not complete
134 any of the two CBCL questionnaires to those who replied to at least one, and we compared
135 characteristics of those who participated at both ages and those who only participated the first time.
136 In both cases we used t-test and Chi²-test to examine differences.

137 To take advantage of the repeated measures of ADHD-symptoms, we analyzed associations with
138 the PFAS using regression models allowing inclusion of the ADHD scores from both age 2½ and 5
139 years into the same model if available and included a unique identification variable to specify
140 related observations (random-effect variable). For these analyses, PFAS concentrations were log₂-
141 transformed.

142 First, we constructed a mixed effect negative binomial regression model for the association between
143 maternal PFAS and the ADHD score, and a standard negative binomial regression model with
144 cluster robust standard errors for the association between child PFAS concentrations and the ADHD
145 score as this model was compliant with the imputed breastfeeding data. Both types of negative
146 binomial regression used the "raw" ADHD scale score as the outcome. The corresponding estimates
147 were incidence-rate ratios (IRR's), which here reflects the relative change in the ADHD scale score
148 with every doubling of the PFAS concentration. Second, the ADHD score was dichotomized to
149 indicate a score ≥ 5 or <5 . A mixed effect logistic regression model was performed while using the
150 maternal PFAS concentrations as exposure indicator, whereas a standard logistic regression with

151 cluster robust standard errors was applied when using the child's PFAS concentrations. For both
152 models, the resultant estimates were odds ratios (OR's) describing the relative change in the odds of
153 scoring equal to or above 5 on the ADHD scale score with every doubling in the PFAS
154 concentration.

155 An interaction term between PFAS and age at CBCL response (2½ years vs 5 years) was included
156 in the regressions to investigate whether the association between PFAS concentration and ADHD
157 symptoms differed at age 2½ years compared to age 5 years. This was done to determine whether
158 we could analyze data from both ages in the same model or if we had to create separate models for
159 the two age groups. For p-values of the interaction term above 0.1, we carried on with the repeated-
160 measure models in which we included ADHD symptoms from both ages. However, if the p-value
161 was < 0.1, results were presented separately for the two ages. Finally, we ran a sensitivity analysis
162 using the non-imputed data on breastfeeding to determine to what extent the results were affected by
163 the breastfeeding-variable used.

164 Covariates to include in the regression models were selected *a priori* using Directed Acyclic Graphs
165 (DAGs) created using DAGitty.net [35]. Relevant factors related to child ADHD symptoms and
166 either maternal PFAS concentrations (Figure 1) or child PFAS concentrations (Figure 2) were
167 included in the DAG. For the model of maternal PFAS concentrations, maternal educational level,
168 parity and parental psychiatric diagnoses were included to control for confounding. For the model
169 of child PFAS concentrations, we included maternal educational level, maternal age at delivery,
170 parity, duration of breastfeeding and parental psychiatric disease. Further, in both models, child sex
171 and age at CBCL response were included as well since they are both strong predictors of the
172 outcome. To investigate any differences according to sex, we included an interaction term between
173 PFAS and child sex. A p-value of the interaction term <0.1 was considered as evidence of sex
174 differences.

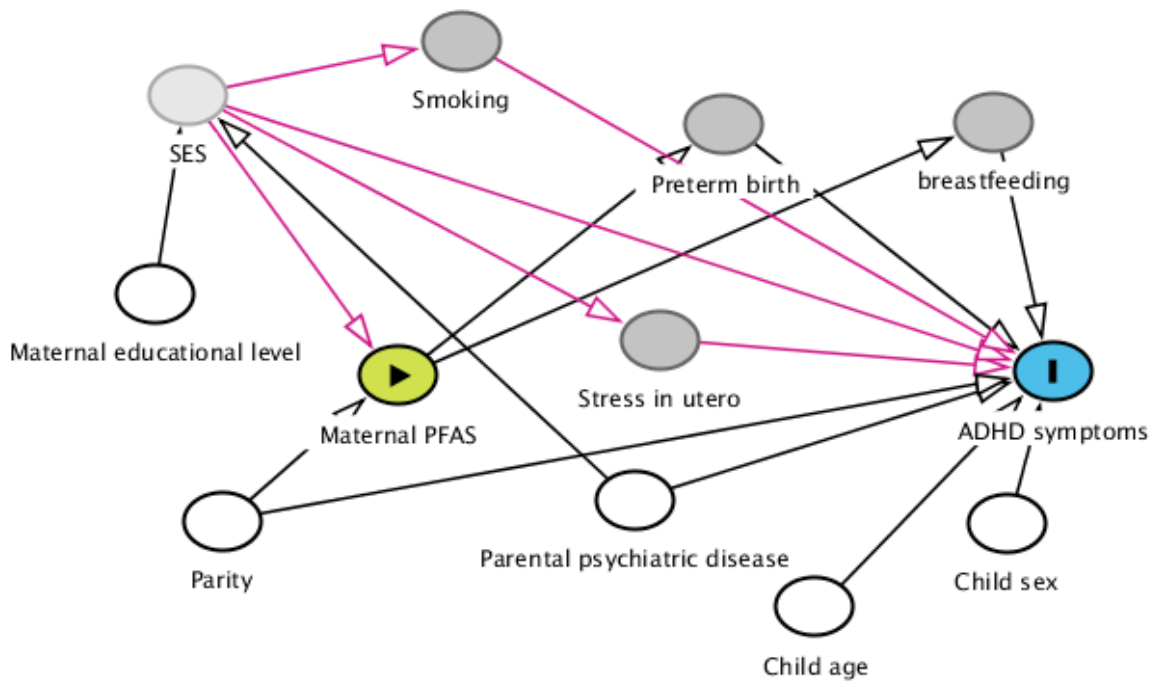


Figure 1: Directed Acyclic Graph (DAG) showing the relationship between maternal PFAS concentration, child ADHD symptoms and related risk factors. Light grey dots: unmeasured variables. White dots: adjusted variables. Black arrows: unbiased paths. Red arrows: biased paths

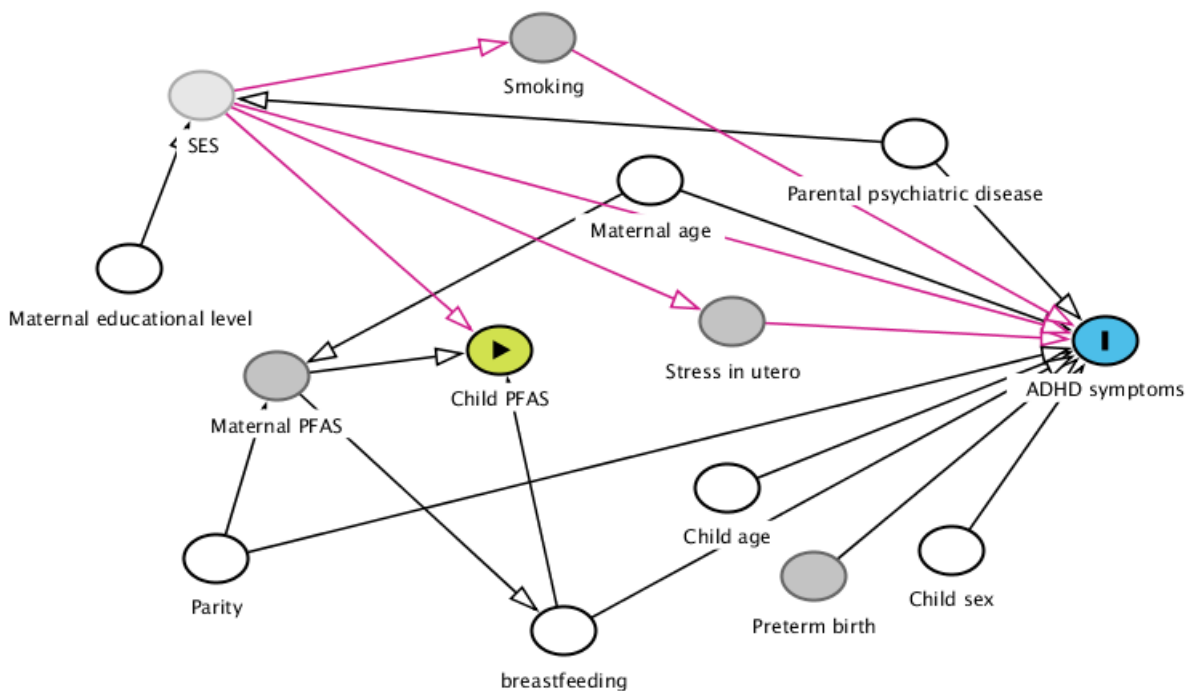
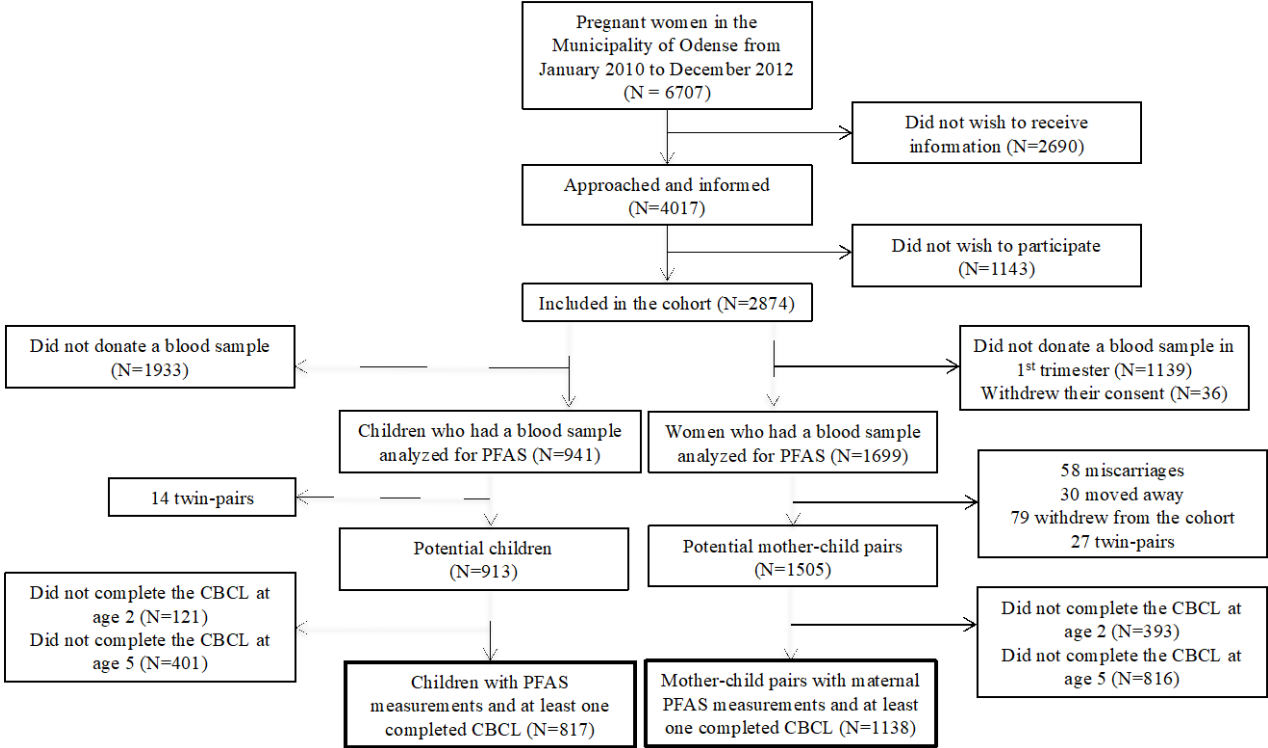


Figure 2: Directed Acyclic Graph (DAG) showing the relationship between maternal PFAS concentration, child ADHD symptoms and related risk factors. Light grey dots: unmeasured variables. White dots: adjusted variables. Black arrows: unbiased paths. Red arrows: biased paths

188 3. Results

189 A total of 1,138 mother-child pairs had a maternal serum-PFAS measurements and a filled-in
190 CBCL from either 2½ years (N=1,112), 5 years (N=689) or both (N= 663). Further, for 817
191 children, a serum-PFAS concentration and a CBCL from 2½ years (N=792), 5 years (N=512) or
192 both (N=487) were available (Figure 3).



194 Figure 3: Flowchart visualizing the selection process of the study participants

195 Finally, 496 mother-child pairs had both maternal and child serum-PFAS concentrations, and at
196 least one CBCL questionnaire completed. Children with two CBCL responses had slightly higher
197 serum concentrations of PFHxS and PFNA at age 18 months (PFHxS (p=0.18) and PFNA
198 (p=0.09)). Further, they generally had older mothers with higher educational level, and slightly
199 higher PFOS (p=0.2), PFNA (p=0.02) and PFDA (p=0.09) concentrations in pregnancy as
200 compared to children with only one CBCL response.

201

Table 1: Child PFAS concentrations at age 18 months (median(p95)) in ng/mL according to characteristics

	N	PFOS	<i>p</i> *	PFOA	<i>p</i>	PFHxS	<i>p</i>	PFNA	<i>p</i>	PFDA	<i>p</i>
Median (95 th percentile)	817	4.65 (11.22)		2.43 (6.40)		0.32 (0.81)		0.58 (1.24)		0.18 (0.37)	
Maternal educational level											
High school, vocational training or less	215	3.85 (10.88)	<0.01	2.04 (6.14)	<0.01	0.26 (0.78)	<0.01	0.50 (1.06)	<0.01	0.16 (0.30)	<0.01
High school + 1-4 years	403	4.93 (11.34)		2.51 (6.74)		0.34 (0.81)		0.59 (1.25)		0.18 (0.36)	
High school + 5 or more years	180	5.20 (11.92)		2.58 (6.27)		0.36 (0.87)		0.63 (1.30)		0.20 (0.43)	
Firstborn											
Yes	443	4.99 (12.28)	<0.01	2.79 (7.21)	<0.01	0.34 (0.88)	<0.01	0.60 (1.31)	0.99	0.18 (0.39)	0.18
No	370	4.30 (10.01)		2.12 (4.26)		0.30 (0.67)		0.57 (1.15)		0.17 (0.35)	
Sex											
Boy	455	4.43 (11.66)	0.79	2.41 (6.33)	0.55	0.31 (0.82)	0.62	0.58 (1.24)	0.77	0.18 (0.36)	0.91
Girl	359	4.92 (11.00)		2.43 (6.66)		0.33 (0.79)		0.58 (1.30)		0.18 (0.39)	
Breastfeeding (weeks)											
0-21	236	2.57 (6.10)	<0.01	1.20 (3.23)	<0.01	0.17 (0.40)	<0.01	0.35 (0.77)	<0.01	0.14 (0.29)	<0.01
22-40	240	5.08 (10.14)		2.73 (6.80)		0.35 (0.71)		0.60 (1.18)		0.18 (0.32)	
41-141	239	6.68 (14.27)		3.11 (7.21)		0.46 (0.99)		0.74 (1.44)		0.21 (0.46)	
Missing	102	5.26 (10.98)		2.45 (6.14)		0.35 (0.83)		0.63 (1.10)		0.20 (0.35)	

*Oneway ANOVA for categorical variables, t-test for binary variables

The median ADHD-scale score was 2 (range 0-12) at both ages 2½ and 5 years, and the 90th percentile was 5 at both ages. Since the ADHD score consists of categoric integers, the exact proportion of children scoring ≥ 5 was 17.4 % at age 2½ years, and 15.8 % at age 5 (Table S2). A higher ADHD scale score was seen in children of women with a lower educational level, who were younger, primiparous, smokers and breastfed for a shorter duration of time. Further, a higher score was seen in boys (≥ 5 : 20.2 at age 2½ and 18.6 at age 5) than in girls (≥ 5 : 14.4 at age 2½ and 12.1 at age 5) (Table S2). The strongest correlations were seen between maternal and child PFOS

212 (r_p : 0.5, $p < 0.01$), maternal and child PFOA (r_p : 0.4, $p < 0.01$), and maternal PFNA and child PFOA (r_p : 0.4, $p < 0.01$). The remaining Pearson
213 correlation coefficients were between 0.1 and 0.4.
214 Higher maternal PFAS concentrations were generally seen in younger women and women with a lower BMI, no previous births and in
215 those with a psychiatric diagnosis (Table S1). Higher child PFAS concentrations were seen with increasing maternal educational level and
216 increasing duration of breastfeeding. Higher concentrations were also observed in firstborns (Table 1).

217

218

219 No interactions between PFAS and age at CBCL response with a p-value <0.1 was observed, and thus, the CBCL scores for age 2½ and 5
 220 years were analyzed together. We found no association between maternal PFAS concentration during pregnancy and child ADHD-scale
 221 score at ages 2½ and 5 years (Table 2).

222

223 Table 2: Incidence rate ratio's (IRR's), odds ratio's (OR's) and 95 % confidence intervals (95% CI) for the association between maternal PFAS concentrations and
 224 child ADHD score

	Crude IRR ^a (95% CI) (N=1138)	Adjusted ^b IRR (95% CI) (N=1113)	Interaction, PFAS and time, p-value	Interaction, PFAS and sex, p-value	Crude OR ^c (95% CI) (N=1138)	Adjusted ^b OR (95% CI) (N=1113)
PFOS	1.01 (0.94, 1.09)	1.00 (0.93, 1.08)	0.25	0.58	0.98 (0.76, 1.27)	0.99 (0.76, 1.29)
PFOA	1.07 (1.01, 1.13)	1.03 (0.97, 1.10)	0.71	0.14	1.09 (0.88, 1.35)	0.97 (0.76, 1.23)
PFHxS	0.99 (0.94, 1.04)	0.98 (0.93, 1.03)	0.96	0.59	0.96 (0.80, 1.16)	0.95 (0.79, 1.16)
PFNA	1.06 (0.98, 1.14)	1.03 (0.96, 1.12)	0.54	0.79	1.11 (0.84, 1.45)	1.06 (0.80, 1.39)
PFDA	1.03 (0.97, 1.10)	1.02 (0.95, 1.09)	0.49	0.99	1.09 (0.86, 1.39)	1.08 (0.85, 1.37)

225

^a: The relative difference in the ADHD score with every doubling of maternal serum-PFAS concentration.

226

^b: Adjusted for parity, maternal educational level, parental psychiatric diagnosis, child sex

227

^c: The relative difference in the odds of scoring above 5 on the ADHD scale with every doubling of maternal serum PFAS concentration

228

229

230

231 Table 3: Incidence rate ratio's (IRR's), odds ratio's (OR's) and 95 % confidence intervals (95% CI) for the association between child PFAS concentrations and ADHD
 232 scale score

	Crude IRR ^a (95% CI) (N=817)	Adjusted ^b IRR (95% CI) (N=775)	Interaction, PFAS and time, p-value	Interaction, PFAS and sex, p-value	Crude OR ^c (95% CI) (N=817)	Adjusted ^b OR (95% CI) (N=775)
PFOS	0.94 (0.77, 1.14)	0.94 (0.87, 1.02)	0.52	0.50	1.16 (0.63, 2.13)	0.90 (0.70, 1.16)
PFOA	0.98 (0.81, 1.20)	1.00 (0.92, 1.08)	0.48	0.33	1.05 (0.63, 1.75)	1.06 (0.84, 1.34)
PFHxS	0.94 (0.79, 1.12)	0.95 (0.88, 1.04)	0.14	0.25	1.28 (0.74, 2.23)	1.04 (0.79, 1.37)
PFNA	0.98 (0.78, 1.25)	0.99 (0.90, 1.08)	0.12	0.65	1.30 (0.68, 2.48)	1.11 (0.84, 1.46)
PFDA	0.95 (0.71, 1.29)	0.98 (0.88, 1.09)	0.09	0.65	1.19 (0.49, 2.89)	1.06 (0.78, 1.44)

233

^a: The relative difference in the ADHD score with every doubling of child serum-PFAS concentration.

234

^b: Adjusted for maternal educational level, duration of breastfeeding, child sex, paternal psychiatric diagnosis, parity, maternal age

235

^c: The relative difference in the odds of scoring above 5 on the ADHD scale with every doubling of child serum PFAS concentration

236

237 Using both the raw ADHD-scale score and the cut-off point of 5 as an outcome, the estimates were
238 all close to 1. We found no evidence of effect modification by child sex in the association between
239 maternal PFAS exposure and ADHD-scale score, although there was a weak indication of a sex
240 difference in the association between PFOA and ADHD score (p-value for interaction=0.14),
241 equivalent to a IRR of 1.08 (95% CI: 0.99; 1.19) in girls, whereas it was 1.00 (0.92, 1.08) in boys,
242 although confidence intervals were overlapping.

243 Further, we found no overall association between child PFAS concentration at age 18 months and
244 ADHD-scale score at age 2½ and 5 years (Table 3). The IRR's were all close to 1, and although the
245 adjusted OR's tended to be slightly above 1, the confidence intervals were wide. Again, there was
246 no evidence of effect modification by child sex . Repeating the analysis using the non-imputed data
247 on breastfeeding, resulted in a nominal change in estimates (<1%; data not shown).

248

249 4. Discussion

250 In this prospective cohort study, we found no association between maternal serum-PFAS
251 concentrations in early pregnancy or child serum-PFAS concentrations at 18 months, and ADHD
252 symptoms at age 2½ and 5 years. The results did not differ between males and females. Thus,
253 although we had repeated measures of both PFAS exposure and ADHD-symptoms in a relatively
254 large study population, our data do not support the hypothesis that PFAS exposure in early life is
255 associated with ADHD.

256 Our findings are in agreement with findings from several previous cohorts. Three studies, including
257 one Danish, using different standardized parent/teacher questionnaires to assess ADHD, found no
258 association between maternal or umbilical cord serum concentrations of PFAS and ADHD behavior
259 in children [16, 18, 19]. Further, three studies using ADHD diagnosis as outcome, found no
260 association with maternal serum or umbilical cord concentrations of PFAS [13-15]. On the other

261 hand, a Danish study found an association between maternal PFNA concentration and behavioral
262 difficulties at age 7 and 11 years [25]. Further, a US study found an association between maternal
263 PFOS, PFHxS and PFNA, and ADHD-like behavior at age 5 and 8 years [24], whereas another
264 study from the US found positive associations between infant blood concentrations of PFOS and
265 PFOA and behavioral difficulties assessed using the Strengths and Difficulties Questionnaire (SDQ)
266 at age 7 years [22].

267 A meta-analysis including nine European studies with maternal/umbilical cord blood concentrations
268 of PFAS, and ADHD symptoms or diagnosis found no overall associations, but did observe higher
269 estimates in girls as compared to boys, thus indicating a possible effect modification by sex [11].
270 Similarly, an interaction effect of child sex on the association between the PFOA concentration at
271 age 2-8 years and ADHD-like behavior 3-4 years later (age range 6-12 years) was found in a study
272 from the C8 Health Project in the US with adverse associations seen in girls, but not in boys [20].
273 Moreover, a recent study found higher odds of having an ADHD diagnosis before age 13 years in
274 children whose mothers had higher PFOS concentrations in their milk. Sex interaction was
275 observed with a much stronger association seen in girls as compared to boys [21].

276 Finally, two Faroese studies investigated the associations between both maternal and child PFAS
277 concentrations and ADHD-like behavior using the SDQ and the Boston Naming Test (BNT). The
278 first found higher serum-PFAS concentrations at age 5 years associated with a higher score on the
279 total SDQ at age 7 (and some subscales). However, no association was seen between maternal
280 PFAS concentrations and SDQ [17]. The study reported evidence of sex-dimorphic associations in
281 cross-sectional analyses at age 7, where positive associations were seen in girls but not in boys. The
282 second study found that maternal PFOS and PFOA were negatively associated with BNT scores and
283 tended to be positively associated with SDQ scores [23]. In those of the above-mentioned studies
284 where sex-differences were investigated, there are some indications that girls are likely more

285 vulnerable to PFAS exposure than boys and that childhood exposure may be more important for
286 neurobehavioral development than prenatal exposure [11]. Additionally, the age of assessment of
287 ADHD-like behavior seems to be higher in studies finding an association. The inconsistent findings
288 across studies could therefore be explained by inconsistency in the timing of both exposure and
289 outcome assessment, and whether sex dimorphism has been investigated. We took two of these
290 factors into account in the present study by including both maternal and child PFAS concentrations
291 for the same population, and by investigating effect modification by sex. We were, however, unable
292 to identify any statistically significant association, whether using maternal or child PFAS
293 concentrations. We did see a weak indication of effect modification by sex in the association
294 between the maternal PFOA concentration and symptoms of ADHD ($p=0.14$). Different
295 explanations have been proposed to explain the apparent vulnerability in girls, including a possible
296 interaction effect of PFAS on estrogen homeostasis. Moreover, as boys are more prone to ADHD,
297 any harmful effect of PFAS may be dwarfed by other risk factors that are stronger associated to
298 ADHD [11]. However, as the p-value seen for the effect modification by sex in the association
299 between PFOA and symptoms of ADHD was above the determined threshold of 0.1, and as no
300 effect modification was found for the other PFAS, we cannot conclude that there was support for a
301 sex difference in our data. Finally, it should be noted that studies finding positive associations
302 reported much higher PFAS concentrations (Oulhote 2016, PFOA median: 4.06 ng/mL; Stein 2014,
303 PFOA median: 35.1 ng/mL) than were seen in the present study (PFOA, median: 2.43 ng/mL).

304

305 Although we did not find evidence of an association in this study, experimental studies have shown
306 that the developing nervous system is a sensitive target for PFAS exposure. According to *in vivo*
307 studies, rodents exposed to PFOS or PFOA pre- and/or postnatally showed increased motor activity,
308 lack of habituation and hyperactivity [36, 37].

309 The present study has several strengths. First, the sample size is large, and participants are from an
310 ongoing cohort study with long follow-up, thereby allowing repeated measures of the outcome and
311 availability of detailed information on participating mothers and children. Second, we had both
312 maternal pregnancy and child PFAS concentrations that allowed us to investigate both pre- and
313 postnatal exposure in the same cohort.

314 Some limitations should be mentioned. First, of the 1,505 eligible mother-child pairs from the OCC,
315 74 % (N=1,112) completed the first CBCL/1½-5 questionnaire and only 46 % the second one
316 (N=689). The attrition cannot be assumed to be completely random, since the child's development
317 and/or behavior may have affected the parents' likelihood to participate in the study. Mothers who
318 did not reply to any of the two CBCL/1½-5 questionnaires were on average younger, less educated
319 and had lower concentrations of PFOS as compared to mothers who replied to at least one of the
320 questionnaires (Table S3). A similar tendency was seen for children for whom only the first CBCL
321 was completed as they on average had younger, less educated mothers and had slightly lower PFAS
322 concentrations as compared to children for whom both CBCL's were completed (data not shown).

323 As younger maternal age and lower educational level are predictors of a higher ADHD risk in
324 children, it is likely that the average ADHD-scale score at both age 2½ and 5 years reported in this
325 study is somewhat underestimating the true score in the population. Fortunately, the nonresponse
326 seemed to be only weakly associated with PFAS exposure, and we know that the decision to
327 participate was not directly affected by exposure status as the blood results was unknown to the
328 families at the time.

329 Second, the fact that we only had both PFAS concentrations from about half of the mother-child
330 pairs, made it hard to investigate the total PFAS exposure profile. Further, as maternal and child
331 PFAS concentrations are correlated to some degree (r_p : 0.1-0.5), it is difficult to separate the effect
332 of one timepoint from the other. Adjusting for maternal age, parity and breastfeeding in the model

333 of postnatal exposure, closed the paths from child PFAS concentration to ADHD through maternal
334 PFAS concentrations (see DAG, figure 2), thereby giving a sensible estimate of the isolated
335 association between child PFAS concentration and the ADHD scale score. However, as early
336 childhood concentrations are almost solely determined by maternal concentrations and duration of
337 breastfeeding, deciding the most vulnerable time window of exposure is very difficult, and in this
338 case also pointless as we did not find any associations between PFAS exposure and the ADHD
339 scale score.

340 Third, the parent-reported ADHD symptoms may be subject to some misclassification, as parents
341 may have different interpretations and expectations in regard to child behavior. However, as this is
342 likely unrelated to exposure status, which was unknown at the time completion of the CBCL, any
343 misclassification will be non-differential and probably bias the estimates toward the null.

344 Nevertheless, an actual diagnosis of ADHD in older children would have been superior to the self-
345 reported symptoms of ADHD.

346 Finally, results were adjusted for maternal educational level as a proxy for SES, and thus, residual
347 confounding from SES cannot be excluded as previous studies have found income to be the
348 strongest driver of an association between SES and PFAS with higher PFAS concentrations seen
349 with higher income levels (Buekers 2018). However, in this case, lack of detailed adjustment for
350 SES will likely bias the effect estimates towards the null.

351

352

353 5. Conclusion

354 This study investigated the association between maternal and child serum-concentrations of PFHxS,
355 PFOS, PFOA, PFNA and PFDA and symptoms of ADHD at ages 2½ and 5 years in children from
356 the Odense Child Cohort. We found no associations between any of the five PFAS measured in

357 serum during pregnancy or at age 18 months and the CBCL/1½-5 ADHD scale score. Further, we
358 found no evidence of effect modification by child sex at the exposure levels encountered.
359 Altogether, our results do not support the hypothesis that PFAS exposure may increase the risk of
360 ADHD-like behavior in children.

References

1. Lau C, Anitole K, Hodes C, Lai D, Pfahles-Hutchens A, Seed J: **Perfluoroalkyl acids: a review of monitoring and toxicological findings.** *Toxicological sciences : an official journal of the Society of Toxicology* 2007, **99**(2):366-394.
2. EFSA CONTAM Panel: **Risk to human health related to the presence of perfluoroalkyl substances in food.** 2020.
3. Sunderland EM, Hu XC, Dassuncao C, Tokranov AK, Wagner CC, Allen JG: **A review of the pathways of human exposure to poly- and perfluoroalkyl substances (PFASs) and present understanding of health effects.** *Journal of exposure science & environmental epidemiology* 2019, **29**(2):131-147.
4. Olsen GW, Burris JM, Ehresman DJ, Froehlich JW, Seacat AM, Butenhoff JL, Zobel LR: **Half-Life of Serum Elimination of Perfluorooctanesulfonate, Perfluorohexanesulfonate, and Perfluorooctanoate in Retired Fluorochemical Production Workers.** *Environmental Health Perspectives* 2007, **115**(9):1298-1305.
5. Li Y, Fletcher T, Mucs D, Scott K, Lindh CH, Tallving P, Jakobsson K: **Half-lives of PFOS, PFHxS and PFOA after end of exposure to contaminated drinking water.** *Occup Environ Med* 2018, **75**(1):46-51.
6. Needham LL, Grandjean P, Heinzow B, Jørgensen PJ, Nielsen F, Patterson DG, Jr., Sjödin A, Turner WE, Weihe P: **Partition of environmental chemicals between maternal and fetal blood and tissues.** *Environmental science & technology* 2011, **45**(3):1121-1126.
7. Mogensen UB, Grandjean P, Heilmann C, Nielsen F, Weihe P, Budtz-Jørgensen E: **Structural equation modeling of immunotoxicity associated with exposure to perfluorinated alkylates.** *Environmental Health* 2015, **14**(1).
8. Grandjean P, Landrigan PJ: **Neurobehavioural effects of developmental toxicity.** *Lancet Neurol* 2014, **13**(3):330-338.
9. Thapar A, Cooper M: **Attention deficit hyperactivity disorder.** *The Lancet* 2016, **387**(10024):1240-1250.
10. Latimer K, Wilson P, Kemp J, Thompson L, Sim F, Gillberg C, Puckering C, Minnis H: **Disruptive behaviour disorders: a systematic review of environmental antenatal and early years risk factors.** *Child Care Health Dev* 2012, **38**(5):611-628.
11. Forns J, Verner MA, Iszatt N, Nowack N, Bach CC, Vrijheid M, Costa O, Andiaarena A, Sovcikova E, Hoyer BB et al: **Early Life Exposure to Perfluoroalkyl Substances (PFAS) and ADHD: A Meta-Analysis of Nine European Population-Based Studies.** *Environ Health Perspect* 2020, **128**(5):57002.
12. Liew Z, Goudarzi H, Oulhote Y: **Developmental Exposures to Perfluoroalkyl Substances (PFASs): An Update of Associated Health Outcomes.** *Curr Environ Health Rep* 2018, **5**(1):1-19.
13. Ode A, Kallen K, Gustafsson P, Rylander L, Jonsson BA, Olofsson P, Ivarsson SA, Lindh CH, Rignell-Hydbom A: **Fetal exposure to perfluorinated compounds and attention deficit hyperactivity disorder in childhood.** *PLoS One* 2014, **9**(4):e95891.
14. Strom M, Hansen S, Olsen SF, Haug LS, Rantakokko P, Kiviranta H, Halldorsson TI: **Persistent organic pollutants measured in maternal serum and offspring neurodevelopmental outcomes--a prospective study with long-term follow-up.** *Environ Int* 2014, **68**:41-48.

15. Liew Z, Ritz B, von Ehrenstein OS, Bech BH, Nohr EA, Fei C, Bossi R, Henriksen TB, Bonefeld-Jorgensen EC, Olsen J: **Attention deficit/hyperactivity disorder and childhood autism in association with prenatal exposure to perfluoroalkyl substances: a nested case-control study in the Danish National Birth Cohort.** *Environ Health Perspect* 2015, **123**(4):367-373.
16. Fei C, Olsen J: **Prenatal exposure to perfluorinated chemicals and behavioral or coordination problems at age 7 years.** *Environ Health Perspect* 2011, **119**(4):573-578.
17. Oulhote Y, Steuerwald U, Debes F, Weihe P, Grandjean P: **Behavioral difficulties in 7-year old children in relation to developmental exposure to perfluorinated alkyl substances.** *Environ Int* 2016, **97**:237-245.
18. Lien GW, Huang CC, Shiu JS, Chen MH, Hsieh WS, Guo YL, Chen PC: **Perfluoroalkyl substances in cord blood and attention deficit/hyperactivity disorder symptoms in seven-year-old children.** *Chemosphere* 2016, **156**:118-127.
19. Skogheim TS, Villanger GD, Weyde KVF, Engel SM, Suren P, Oie MG, Skogan AH, Biele G, Zeiner P, Overgaard KR et al: **Prenatal exposure to perfluoroalkyl substances and associations with symptoms of attention-deficit/hyperactivity disorder and cognitive functions in preschool children.** *Int J Hyg Environ Health* 2020, **223**(1):80-92.
20. Stein CR, Savitz DA, Bellinger DC: **Perfluorooctanoate exposure in a highly exposed community and parent and teacher reports of behaviour in 6-12-year-old children.** *Paediatr Perinat Epidemiol* 2014, **28**(2):146-156.
21. Lenters V, Iszatt N, Forns J, Cechova E, Kocan A, Legler J, Leonards P, Stigum H, Eggesbo M: **Early-life exposure to persistent organic pollutants (OCPs, PBDEs, PCBs, PFASs) and attention-deficit/hyperactivity disorder: A multi-pollutant analysis of a Norwegian birth cohort.** *Environ Int* 2019, **125**:33-42.
22. Ghassabian A, Bell EM, Ma WL, Sundaram R, Kannan K, Buck Louis GM, Yeung E: **Concentrations of perfluoroalkyl substances and bisphenol A in newborn dried blood spots and the association with child behavior.** *Environ Pollut* 2018, **243**(Pt B):1629-1636.
23. Oulhote Y, Coull B, Bind MA, Debes F, Nielsen F, Tamayo I, Weihe P, Grandjean P: **Joint and independent neurotoxic effects of early life exposures to a chemical mixture: A multi-pollutant approach combining ensemble learning and g-computation.** *Environ Epidemiol* 2019, **3**(5).
24. Vuong AM, Webster GM, Yolton K, Calafat AM, Muckle G, Lanphear BP, Chen A: **Prenatal exposure to per- and polyfluoroalkyl substances (PFAS) and neurobehavior in US children through 8 years of age: The HOME study.** *Environ Res* 2021, **195**:110825.
25. Luo J, Xiao J, Gao Y, Ramlau-Hansen CH, Toft G, Li J, Obel C, Andersen SL, Deziel NC, Tseng WL et al: **Prenatal exposure to perfluoroalkyl substances and behavioral difficulties in childhood at 7 and 11 years.** *Environ Res* 2020, **191**:110111.
26. Kyhl HB, Jensen TK, Barington T, Buhl S, Norberg LA, Jorgensen JS, Jensen DF, Christesen HT, Lamont RF, Husby S: **The Odense Child Cohort: aims, design, and cohort profile.** *Paediatr Perinat Epidemiol* 2015, **29**(3):250-258.
27. Dalsager L, Fage-Larsen B, Bilenberg N, Jensen TK, Nielsen F, Kyhl HB, Grandjean P, Andersen HR: **Maternal urinary concentrations of pyrethroid and chlorpyrifos metabolites and attention deficit hyperactivity disorder (ADHD) symptoms in 2-4-year-old children from the Odense Child Cohort.** *Environ Res* 2019, **176**:108533.
28. Vorkamp K, Nielsen F, Kyhl HB, Husby S, Nielsen LB, Barington T, Andersson AM, Jensen TK: **Polybrominated diphenyl ethers and perfluoroalkyl substances in serum of pregnant**

- women: levels, correlations, and potential health implications. *Arch Environ Contam Toxicol* 2014, **67**(1):9-20.
29. Bilenberg N: **The Child Behavior Checklist (CBCL) and related material: standardization and validation in Danish population based and clinically based samples.** *Acta Psychiatr Scand Suppl* 1999, **398**:2-52.
 30. Kristensen S, Henriksen TB, Bilenberg N: **The Child Behavior Checklist for Ages 1.5-5 (CBCL/1(1/2)-5): assessment and analysis of parent- and caregiver-reported problems in a population-based sample of Danish preschool children.** *Nordic journal of psychiatry* 2010, **64**(3):203-209.
 31. Achenbach TM, & Rescorla, L. A. (2000). Burlington, VT: University of Vermont, Research Center for Children, Youth, & Families.: **Manual for the ASEBA Preschool Forms & Profiles.** 2000.
 32. Doyle A, Ostrander R, Skare S, Crosby RD, August GJ: **Convergent and criterion-related validity of the Behavior Assessment System for Children-Parent Rating Scale.** *Journal of clinical child psychology* 1997, **26**(3):276-284.
 33. Kerr DC, Lunkenheimer ES, Olson SL: **Assessment of child problem behaviors by multiple informants: a longitudinal study from preschool to school entry.** *J Child Psychol Psychiatry* 2007, **48**(10):967-975.
 34. Aebi M, Winkler Metzke C, Steinhausen HC: **Accuracy of the DSM-oriented attention problem scale of the child behavior checklist in diagnosing attention-deficit hyperactivity disorder.** *J Atten Disord* 2010, **13**(5):454-463.
 35. Textor J, Hardt J, Knapp S: **DAGitty: a graphical tool for analyzing causal diagrams.** *Epidemiology (Cambridge, Mass)* 2011, **22**(5):745.
 36. Johansson N, Fredriksson A, Eriksson P: **Neonatal exposure to perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA) causes neurobehavioural defects in adult mice.** *Neurotoxicology* 2008, **29**(1):160-169.
 37. Onishchenko N, Fischer C, Wan Ibrahim WN, Negri S, Spulber S, Cottica D, Ceccatelli S: **Prenatal exposure to PFOS or PFOA alters motor function in mice in a sex-related manner.** *Neurotox Res* 2011, **19**(3):452-461.